

Kenny Nguyen

US Citizen - Gilbert, Arizona

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Education

- MS** **Arizona State University**, MS in Computer Science Aug 2026 – Dec 2027
- Accelerated 4+1 Program
- BS** **Arizona State University**, BS in Computer Science Aug 2022 – May 2026
- GPA: 3.88/4.0
 - Relevant Coursework: Foundations of Algorithms, Operating Systems, Computer Networks, Data Structures and Algorithms, Software QA and Testing, Principles of Programming Languages

Experience

- Tokyo Electron America**, Software Engineering Intern — Senior Capstone Aug 2025 – May 2026
- Built a C# Unity state machine for NPC behavior and event-driven communication between subsystems, enabling state changes to trigger coordinated behavior across the application.
 - Integrated independently developed Unity modules into a single Meta Quest VR application, establishing reliable initialization, scene transitions, and end conditions for standalone Android deployment.
 - Resolved integration failures that emerged only in combined runtime, including geometry-collision and UI-layout issues across Quest 2 and Quest 3 hardware.
 - Produced integration and test handoff documentation for client engineers and presented the completed application to academic and industry stakeholders.

Projects

- Redis Server in Go | Go, TCP, RESP, Concurrency, Persistence, Pub/Sub** GitHub
- Built a Redis-inspired server in Go, implementing 13 commands over TCP with a buffered RESP parser for persistent client connections.
 - Designed a mutex-protected in-memory store supporting strings with TTL expiration and Redis Streams, including XADD, XRANGE, and blocking XREAD.
 - Implemented concurrent Pub/Sub delivery with goroutine-per-connection handling and a synchronized channel-to-subscriber registry.
 - Added AOF persistence for SET operations with RESP-encoded write logging, configurable synchronous disk writes, manifest-selected AOF recovery, and command replay on startup; validated behavior with Go unit and CodeCrafters integration tests.
- Distributed Storage System | Python, UDP Sockets, Multithreading**
- Built a 3-process distributed storage system with a manager, client, and multithreaded disk nodes communicating over UDP through a custom 23-command protocol.
 - Implemented rotating XOR parity across N disks to recover from a single disk failure, reconstructing lost data from surviving blocks with parallel threaded I/O.
 - Designed protocol handlers for storage-node coordination, user registration, file ownership enforcement, and recovery workflows.

Skills

Languages: Python, Go, Java, C/C++, JavaScript, TypeScript, HTML/CSS, Bash, SQL

Frameworks & Databases: Spring Boot, React, JUnit, Mockito, PostgreSQL

Tools & Concepts: Kafka, Docker, Git, Linux/Unix, Cursor, Claude Code, REST APIs, Microservices